

what context is the RAIL concept applicable? Can you give an example?

8. Consider this code snippet:

```
class Empty {};
Empty e1;

int main(void)
{
    printf("size of empty class is %u\n", sizeof(e1));
}
```

What is the output of this code snippet? If size of the empty class as printed out by the function above is not zero, why is it so?

9. Consider a C++ code snippet with two classes, a base class BASE and a derived class DERIVED, which is derived from BASE:

```
BASE b;
DERIVED d;
b=d;
```

Do you see any issues in the assignment statement "b = d"?

10. Now consider this code snippet:

```
BASE* b = new BASE();
DERIVED* d = new DERIVED();
b=d;
```

What is the issue with the above code snippet?

11. Here is a problem from the book *Algorithms* by Carmen, Leicerson, Rivest and Stein ('The CLRS book', as it is popularly known), which is quite often asked in software interviews. Let's suppose that you are given n red and n blue water jugs, all of different shapes and sizes. All red jugs hold different amounts of water, as do the blue ones. Moreover, for every red jug, there is a blue jug that holds the same amount of water, and vice-versa. It is your task to group the jugs into pairs of red and blue jugs that hold the same amount of water. To do so, you may perform the following operation: pick a pair of jugs in which one is red and one is blue, fill the red jug with water, and then pour the water into the blue jug. This operation will tell you whether the red or the blue jug can hold more water, or if they are of the same volume. Assume that such a comparison takes one time unit. Your goal is to find an algorithm that makes a minimum number of comparisons to determine the grouping. Note that you are *not* allowed to directly test a red jug against a red jug, or a blue jug against a blue jug.

I would suggest that if you are appearing for any programming interviews, it would be good to refresh your memory on the basic algorithms from the CLRS book. Working out the exercises from the book is a good way to prepare for the interview.

12. You know that *new* and *delete* are the C++ memory allocation functions similar to *malloc* and *free* in C. What is the equivalent of *realloc* in C++?

13. Given an undirected graph $G(V, E)$, where V represents the set of vertices and E represents the set of edges, and two colours red and green, can you come up with an algorithm to determine: (a) whether the graph is colourable with the two colours; and (b) if it is colourable, give an algorithm to colour the graph. (Note that a proper colouring requires that no two adjacent vertices are allocated the same colour and two vertices are adjacent if there is an edge between them.)

14. Consider the following code snippet:

```
//MyString is a string class I have defined and not the STL string class
const MyString my_str("This is Sandya's string, not a std::string");
for(int i(0); i < my_str.length(); ++i)
{
    // Do something with my_str[i]
}
```

What is the time complexity of the above loop? Is there any C++ language feature you can use to improve the complexity? (Assume that the 'length()' routine has a complexity of $O(N)$ where N is the length of the string.)

15. Consider the following code snippet:

```
string foo() { return "sandyas"; }
void bar() {
    const string& s1 = foo();
    cout << s1 << endl;
}
```

Do you see any issues with the above code? *Hint: Remember that a temporary object lasts only until the end of the full expression in which it appears.* If yes, what is the issue? If not, state the reason.

16. You are given an array of N integers, which may contain duplicates. You are also given a value ' k '. You are asked to find if there are two integers in the array whose sum is equal to k . Note that the array contains both positive and negative elements. What is the complexity of your algorithm? How would it change if you are told to determine if there are three integers in the array whose sum is equal to ' k '.

I would like to request readers to send their solutions to these questions. Also, if you are preparing for computer science programming interviews, you may find it useful to visit my programming interviews discussion group 'Computer Science Interview Training (India)' on LinkedIn (www.linkedin.com).

If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me along with your solutions and feedback at sandyasm_AT_yahoo_DOT_com. Till we meet again next month, wish you a merry Christmas and a happy new year! 

About the author:

Sandya Mannarswamy. The author is a specialist in compiler optimisation and works at Hewlett-Packard India. She has a number of publications and patents to her credit, and her areas of interest include virtualisation technologies and software development tools.